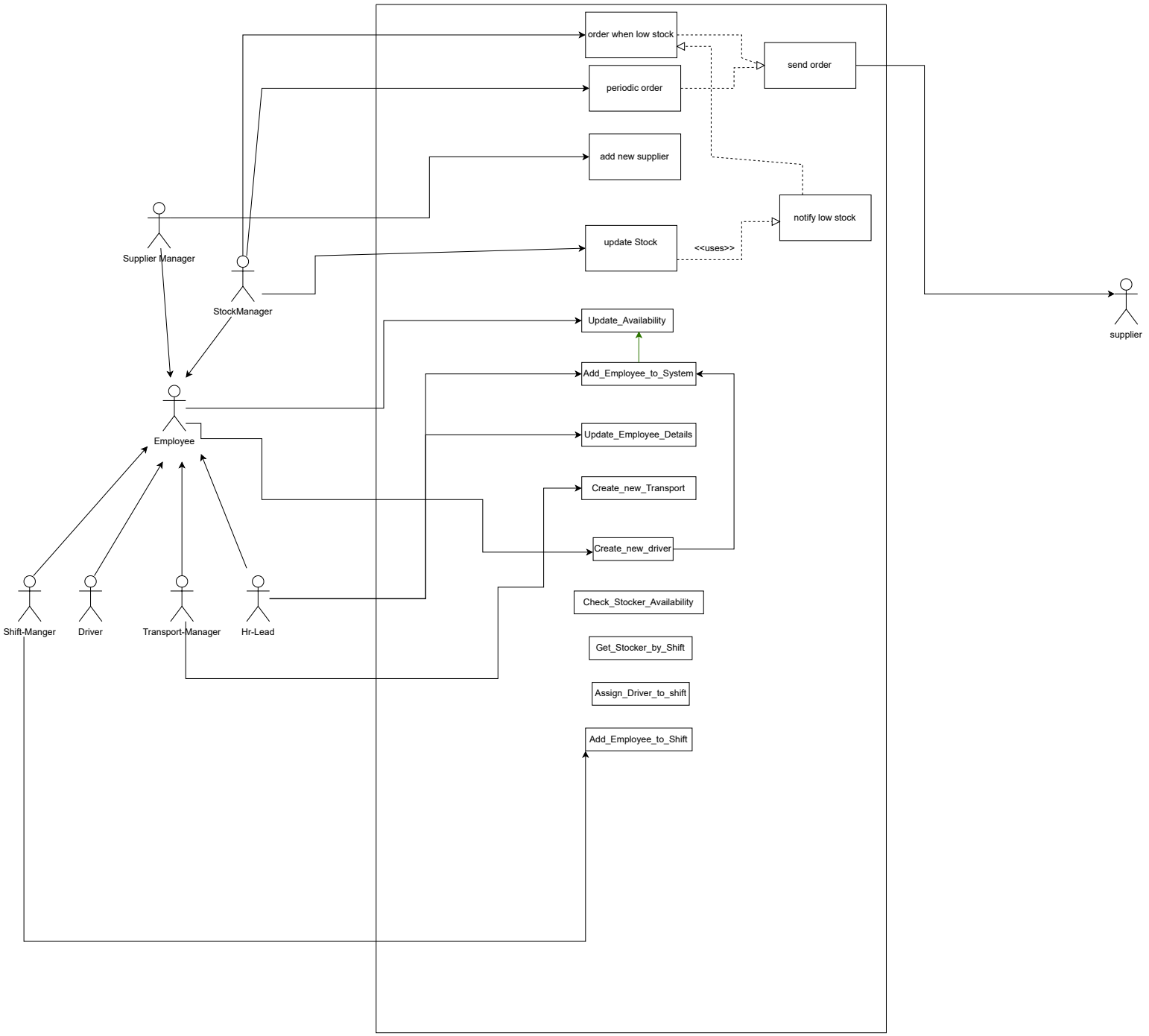

order when low stock
send order
periodic order
add new supplier
notify low stock
Supplier Manager
update Stock
<<uses>>
StockManager
supplier
Update_Availability
Add_Employee_to_System
Employee
Update_Employee_Details
Create_new_Transport
Create_new_driver
Shift-Manger
Driver
Transport-Manager
Hr-Lead
Check_Stocker_Availability
Get_Stocker_by_Shift
Assign_Driver_to_shift
Add_Employee_to_Shift

Use case name:
Add employee to shift

Textual Description:
The process begins with the manager providing input to specify the desired shift ID, the employee ID of the worker to be added, and the role the employee will take on within that shift.
If all inputs meet the necessary conditions, the employee is successfully assigned to the shift with the designated role. Otherwise, appropriate error messages are displayed to notify the manager.

List of Actors:
Shift Manager

Pre-conditions:
The shift ID entered must correspond to an existing shift in the ShiftsDB, the employee ID must correspond to an existing employee in the EmployeesDB, the shift being assigned must have a designated Shift Manager, the employee must be available for the specific type of shift being worked with, the employee must possess the required role for the shift.

Post-conditions:
a copy of the specified employee object is added to the assigned employees list within the corresponding shift object.

Main success scenario:

PRINT "Enter the Employee ID you want to assign to a shift:"
READ employeeID from manager input
employee ← findEmployeeById(employee ID)

PRINT "Enter the Shift ID to assign the employee to:"
READ shiftID from input
shift ← findShiftById(shift ID)

PRINT "Select the corresponding number to the desired Role to add:"
CALL displayAvailableRoles()
selectedRole ← promptForRole(choice)

IF succeeded to add employee to shift THEN
PRINT "Employee <employee name> added to shift <shift ID> with role <roleName>"

Alternatives/Extensions:

In case of a failure – we display the specific reason for failure.

IF employee does not exist THEN
    PRINT "Employee with ID <employee ID> does not exist."
    RETURN

```
IF shift does not exist THEN
    PRINT "Shift with ID <shift ID> does not exist."
    RETURN

isAvailable ← checkEmployeeAvailability(employee, shift.date)
IF NOT isAvailable THEN
PRINT "Employee <employee name> is not available for this shift date."
RETURN

IF employee already assigned to shift THEN
     PRINT "Employee <employee name> is already assigned to this shift."
      RETURN

IF employee NOT has selected role THEN
      PRINT "Employee <employee name> does not have the role <roleName>"
      RETURN

IF NOT succeeded to add employee to shift THEN
     PRINT "Failed to add employee to the shift."
```

contract:

Operation:
addEmployeeToShift()

Cross References:
Use Case: Add employee to shift

Preconditions:

- The specified shift id exists in the ShiftsDB.
- The specified employee id exists in the EmployeesDB.
- The shift has an assigned Shift Manager.
- The employee is available for the type of shift being assigned.
- The employee possesses the required role for the shift.

Postconditions:

- A copy of the specified employee object was added to the assigned employees list within the corresponding shift object.
- A success message is displayed if the employee is successfully added.

Use case name: Create a transport
Textual Description: The system gets an items list, a branch and a designated delivery date and time. The system creates transport which consists of transport routes from supplier(s) to branch. We attach a truck and a relevant driver.
List of Actors: Transport Manager, Driver.
Pre-conditions: There exists a Stocker at the branch at the given delivery date-time. There exists a suitable supplier for at least part of the required items. There exists a relevant truck for the delivery at the required time. And there exists a driver for the relevant time frame with a suitable license type.
Post-conditions: We create a transport with association to transport routes which associates to a delivery file, the transport also associates with a truck and a driver.
Main success scenario: INPUT: itemsList, branchId, deliveryDateTime 1. VERIFY there is a Stocker available at the branch at deliveryDateTime 2. FIND suppliers who can supply at least part of the items 3. IF no suitable supplier found → FAIL 4. FIND a truck that is available at deliveryDateTime 5. IF no truck found → FAIL 6. FIND a driver who: - is available at deliveryDateTime - has a valid license for the truck 7. IF no suitable driver found → FAIL 8. CREATE Transport entity 9. FOR each supplier: a. CREATE TransportRoute from supplier to branch b. CREATE DeliveryFile for route c. ASSOCIATE DeliveryFile with TransportRoute d. ADD TransportRoute to Transport 10. ASSOCIATE truck and driver with the Transport 11. SAVE Transport to database OUTPUT: Transport created successfully

Alternatives/Extensions:

IF suppliers.isEmpty():

DISPLAY "No supplier found for required items"

ABORT transport creation

IF truck == null:

DISPLAY "No truck available at the required time"

ABORT transport creation

IF driver == null:

DISPLAY "No driver available or valid for selected truck"

ABORT transport creation

IF only some items can be fulfilled:

CREATE transport routes for available items

FLAG transport as "Partially fulfilled"

IF truck or driver already assigned to another transport at same time:

DISPLAY "Truck or driver already in use"

ALLOW user to select different resources or reschedule

Contract:

Operation: createTransport(itemsList : List<Item>, branchId : int, deliveryDateTime : datetime)

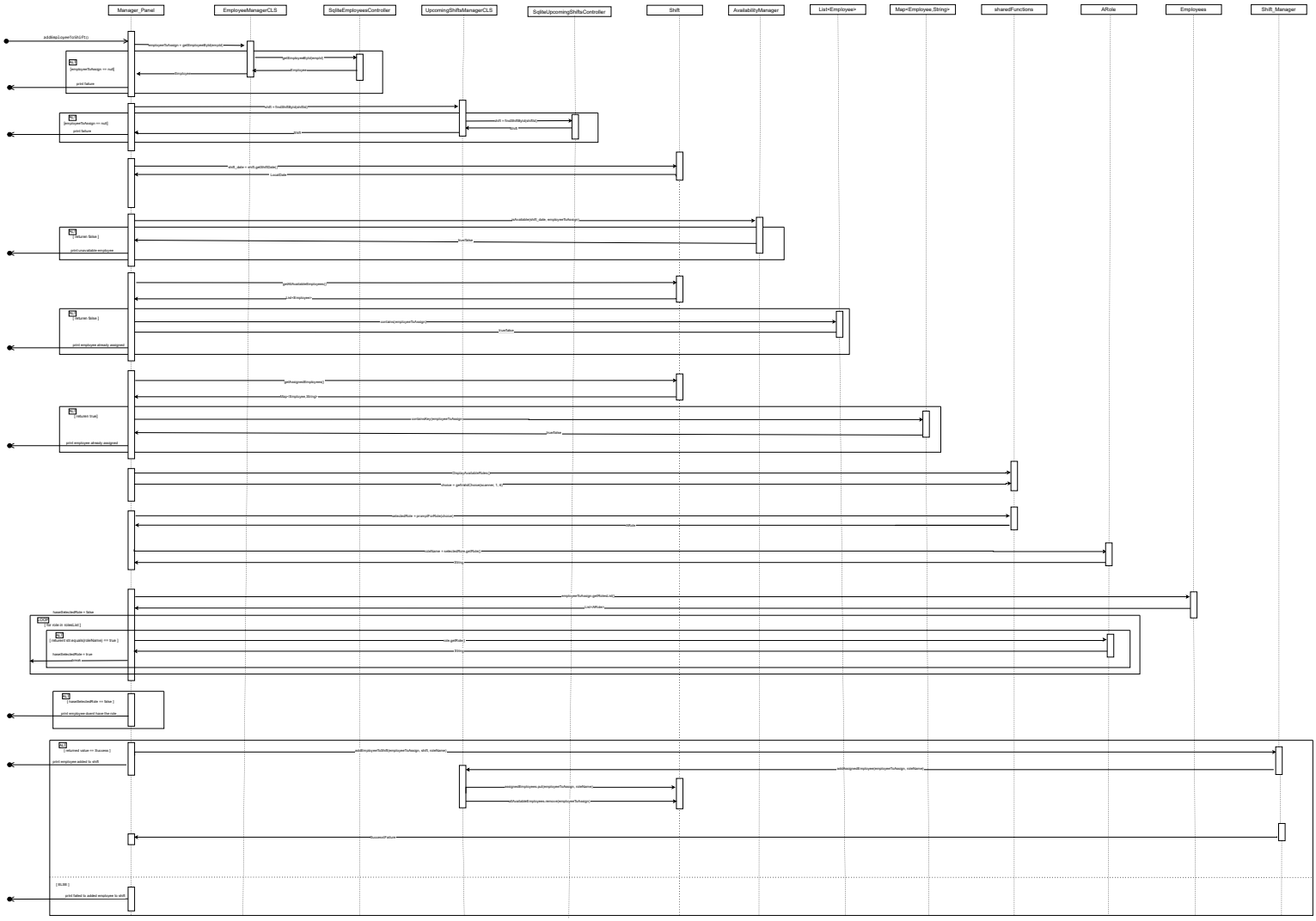
References:
Use Case: Create a Transport

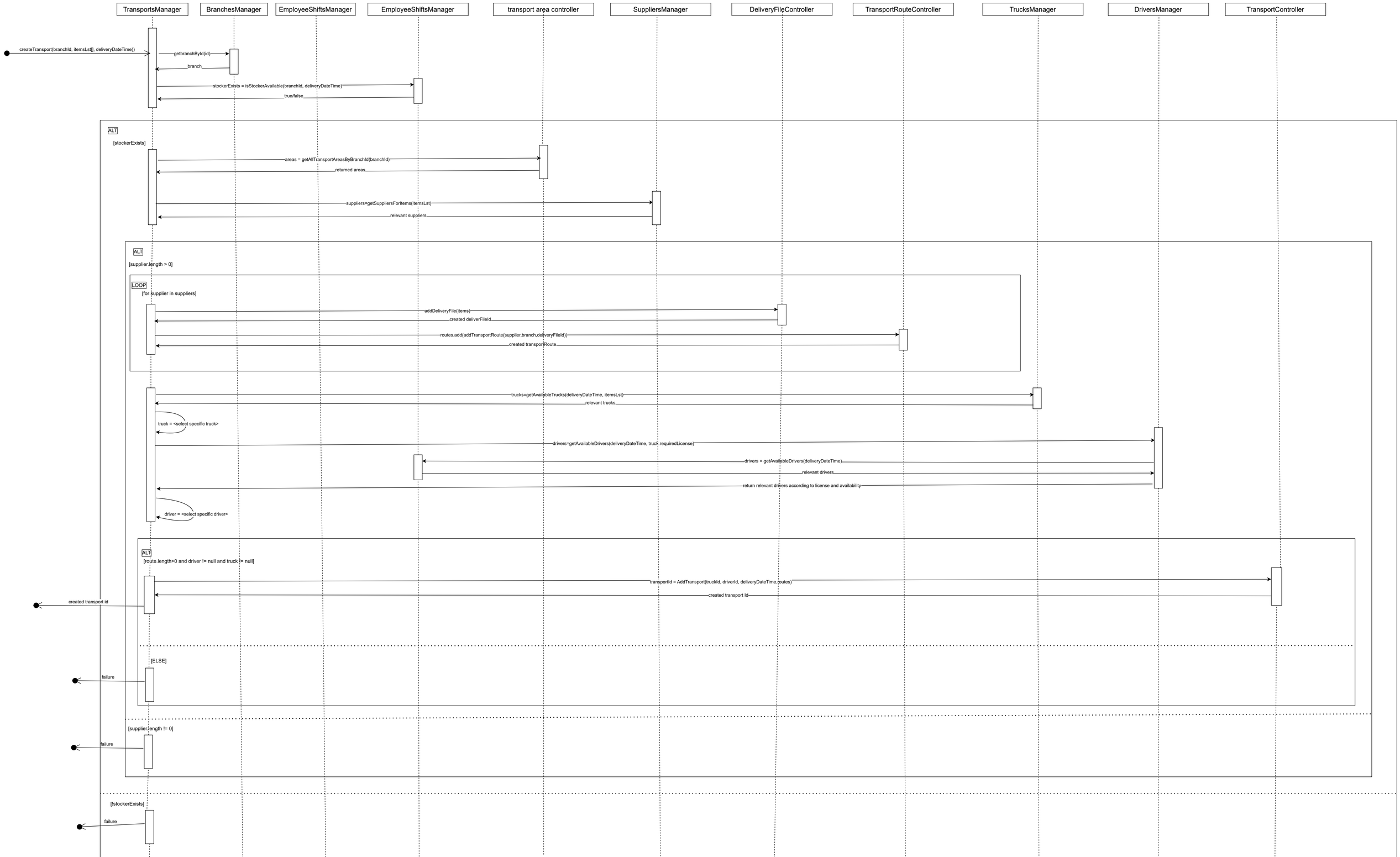
Preconditions:

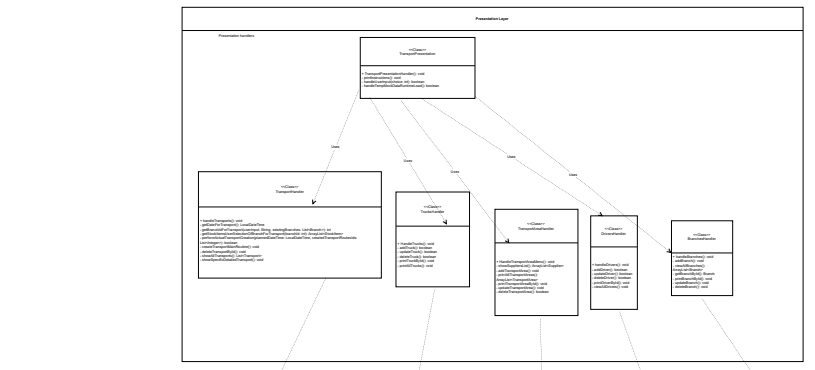
- There is a Stocker available at the specified branch at deliveryDateTime.
- There exists at least one supplier that can supply part of the requested items.
- There is a truck available for delivery at the required time.
- There is a driver available during the time window with a valid license for the selected truck.

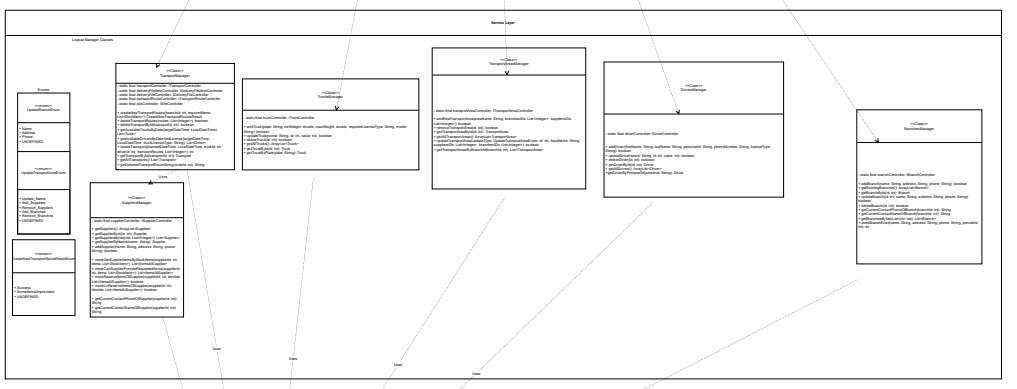
Postconditions:

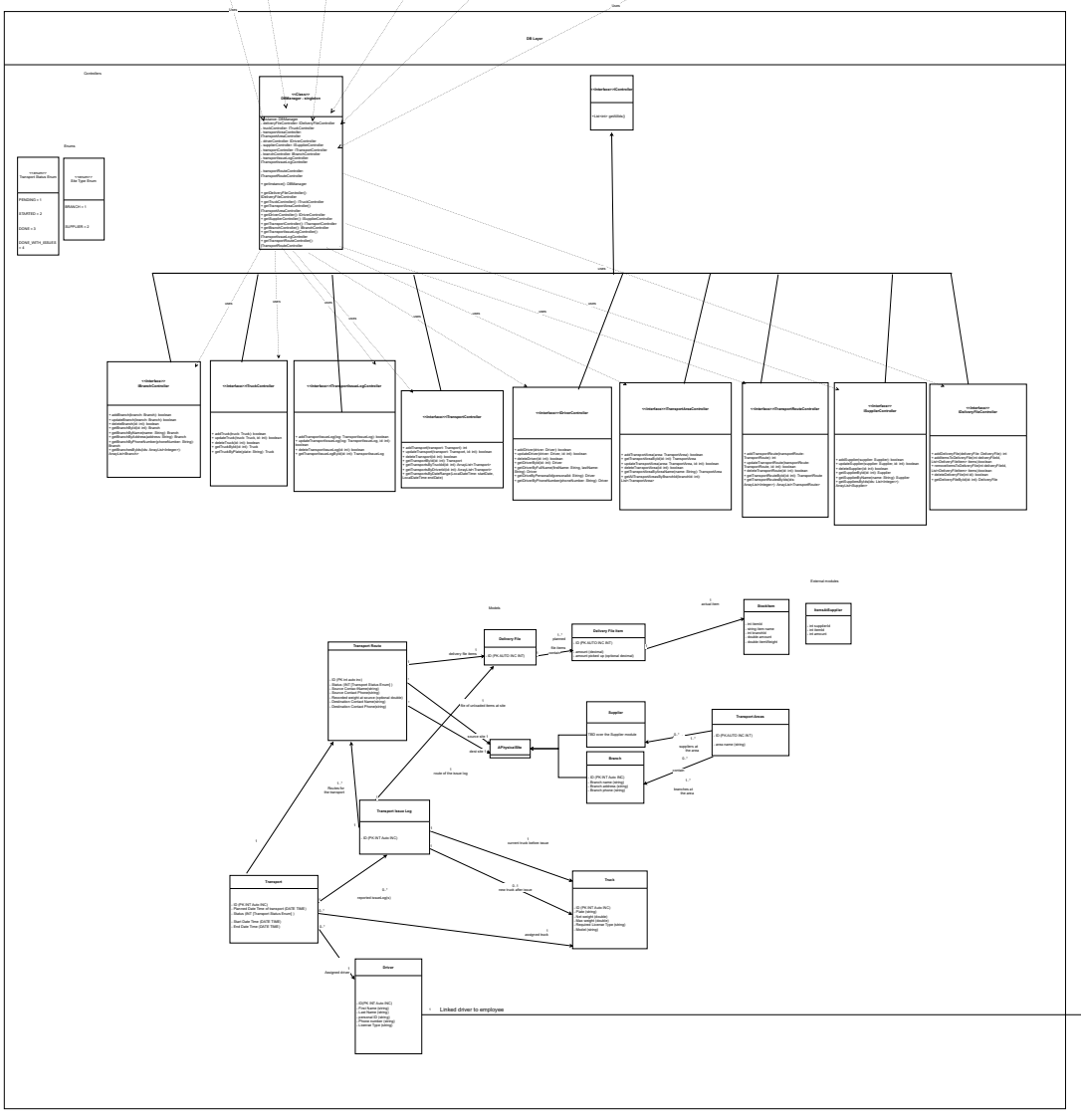
- A Transport instance was **created**.
- The Transport was **associated** with one or more TransportRoute instances.
- Each TransportRoute was **associated** with a DeliveryFile.
- The Transport was **associated** with a selected Truck.
- The Transport was **associated** with a selected Driver.

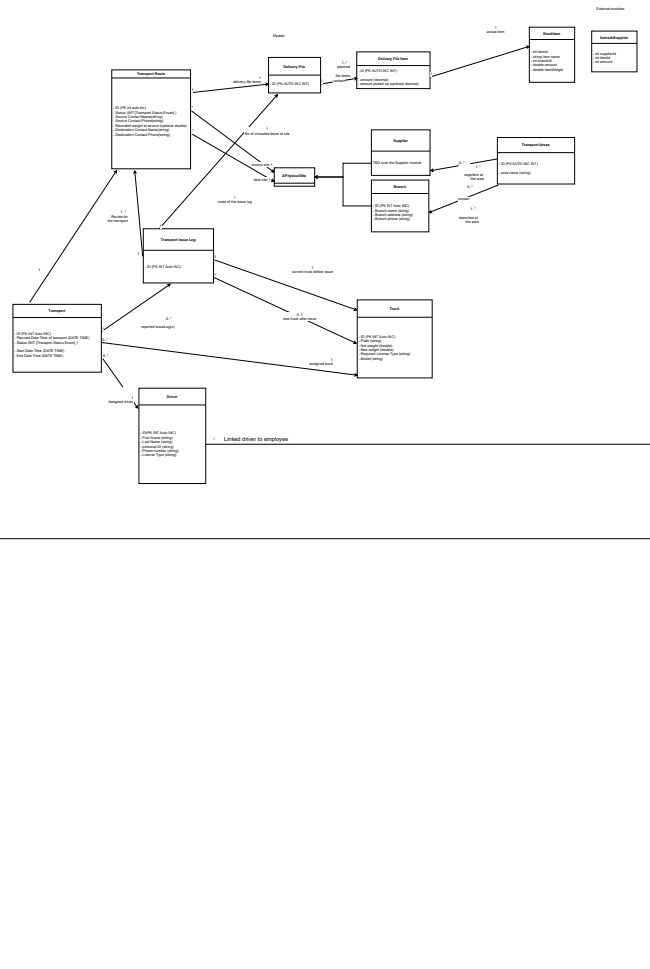

Manager_Panel
EmployeesManagerCLS
SqliteEmployeesController
UpcomingShiftsManagerCLS
SqliteUpcomingShiftsController
Shift
AvailabilityManager
List<Employee>
Map<Employee,String>
sharedFunctions
ARole
Employees
Shift_Manager
addStep2LayerToVbox()
employeeToAssign = getEmployeeById(empId)
getEmployeeById(empId)
Employee
String(empId)
print failure
[employeeToAssign == null]
shift = fromStrToShift(shiftStr)
shift
shift_date = shift.getShiftDate()
LocalDate
isAvailable(shift_date, employeeToAssign)
boolValue
print unavailable employee
getAllAvailableEmployees()
List<Employee>
[isReturn false]
contains(employeeToAssign)
boolValue
print employee already assigned
getAssignedEmployees()
Map<Employee,String>
containsKey(employeeToAssign)
boolValue
[isReturn true]
displayAvailableRoles()
choice = getValidChoice(scanner, 1, 4)
selectedRole = parseRoleByChoice(choice)
ARole
roleName = selectedRole.getRole()
String
employeeToAssign.getRoles().set()
hasValidRole = false
loop
[i < roles.length && !hasValid]
role_getRole()
String
hasValidRole = true
i++
[hasValidRole == false]
print employee doesnt have the role
[returned value == true/area]
addEmployeeToShift(employeeToAssign, shift, roleName)
addAssignedEmployee(employeeToAssign, roleName)
assignedEmployees.put(employeeToAssign, roleName)
availableEmployees.remove(employeeToAssign)
saveToFile()
[ELSE]
print failed to added employee to shift


TransportsManager
BranchesManager
EmployeeShiftsManager
EmployeeShiftsManager
transport area controller
SuppliersManager
DeliveryFileController
TransportRouteController
TrucksManager
DriversManager
TransportController
createTransport(branchId, itemsLst[], deliveryDateTime))
getbranchById(id)
branch
stockerExists = isStockerAvailable(branchId, deliveryDateTime)
true/false
ALT
[stockerExists]
areas = getAllTransportAreasByBranchId(branchId)
returned areas
suppliers=getSuppliersForItems(itemsLst)
relevant suppliers
ALT
[supplier.length > 0]
LOOP
[for supplier in suppliers]
addDeliveryFile(items)
created deliverFileId
routes.add(addTransportRoute(supplier,branch,deliveryFileId))
created transportRoute
trucks=getAvailableTrucks(deliveryDateTime, itemsLst)
relevant trucks
truck = <select specific truck>
drivers=getAvailableDrivers(deliveryDateTime, truck.requiredLicense)
drivers = getAvailableDrivers(deliveryDateTime)
relevant drivers
return relevant drivers according to license and availability
driver = <select specific driver>
ALT
[route.length>0 and driver != null and truck != null]
transportId = AddTransport(truckId, driverId, deliveryDateTime,routes)
created transport Id
created transport id
[ELSE]
failure
[supplier.length != 0]
failure
[!stockerExists]
failure

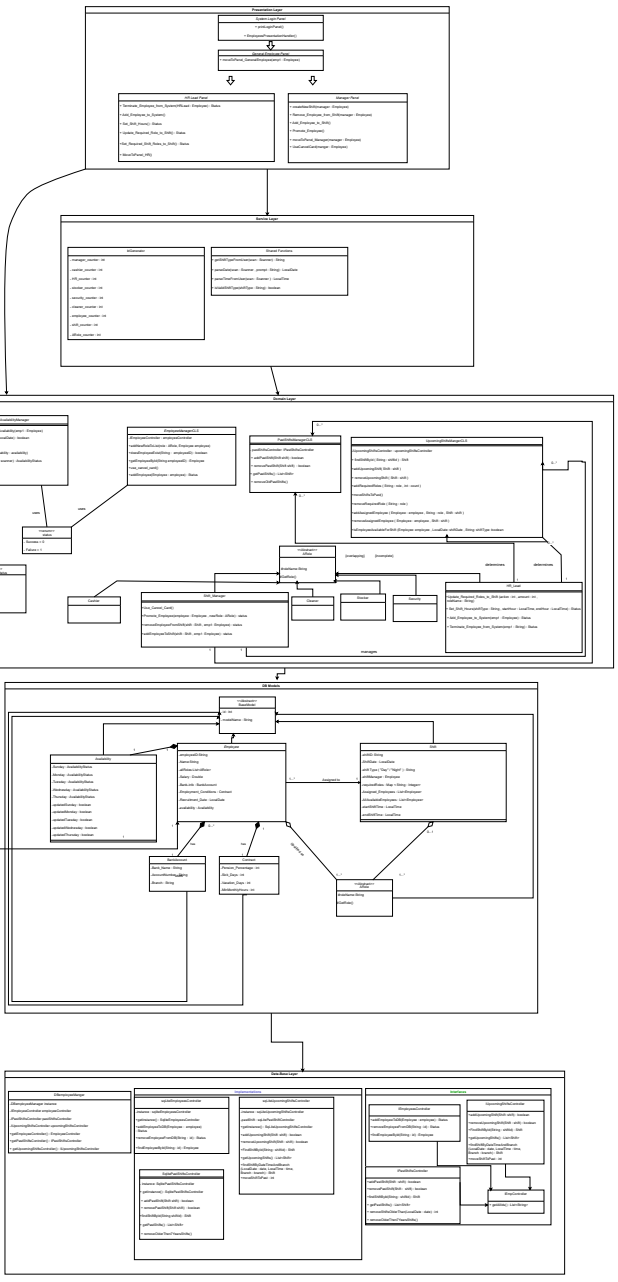


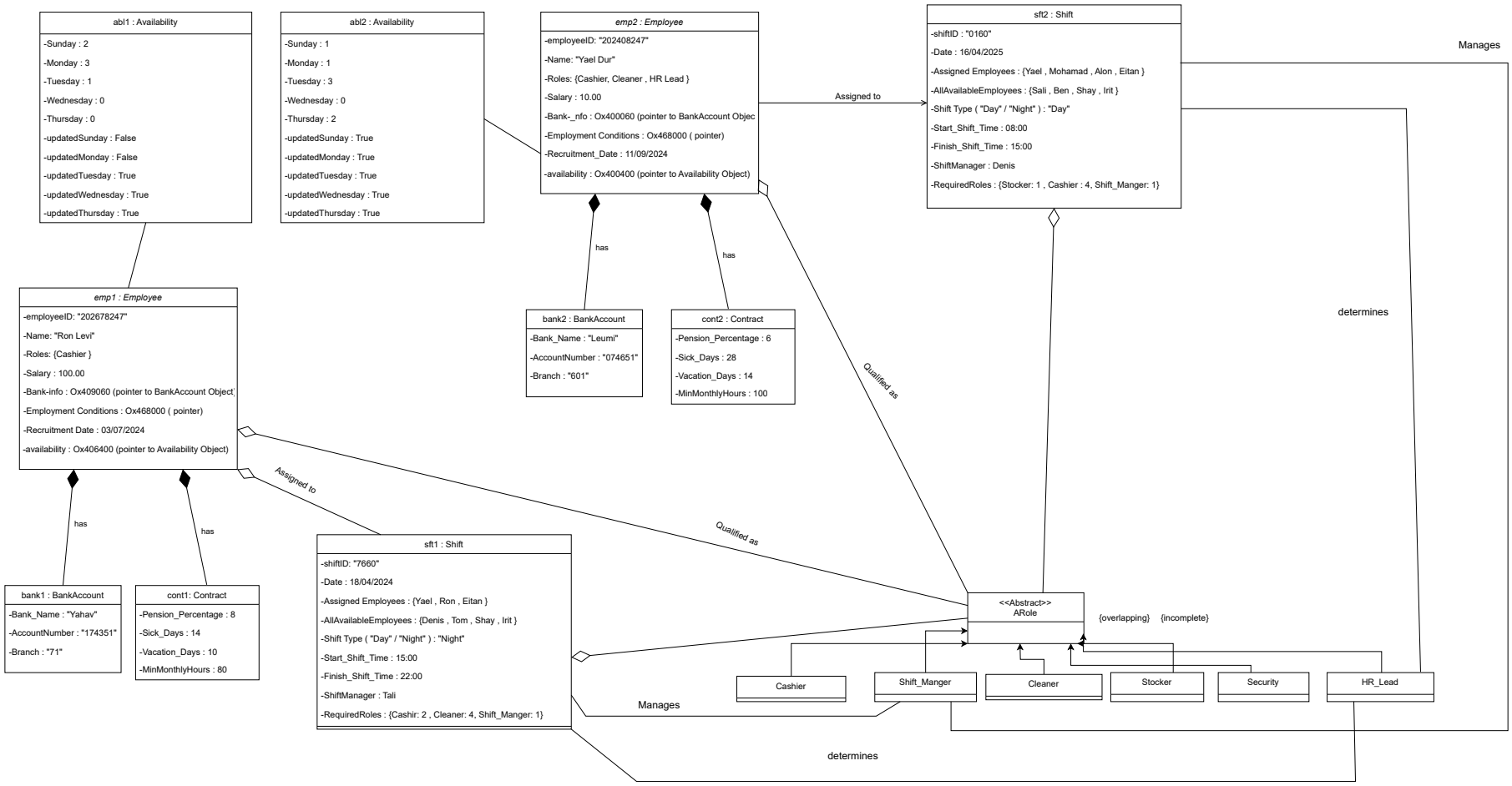


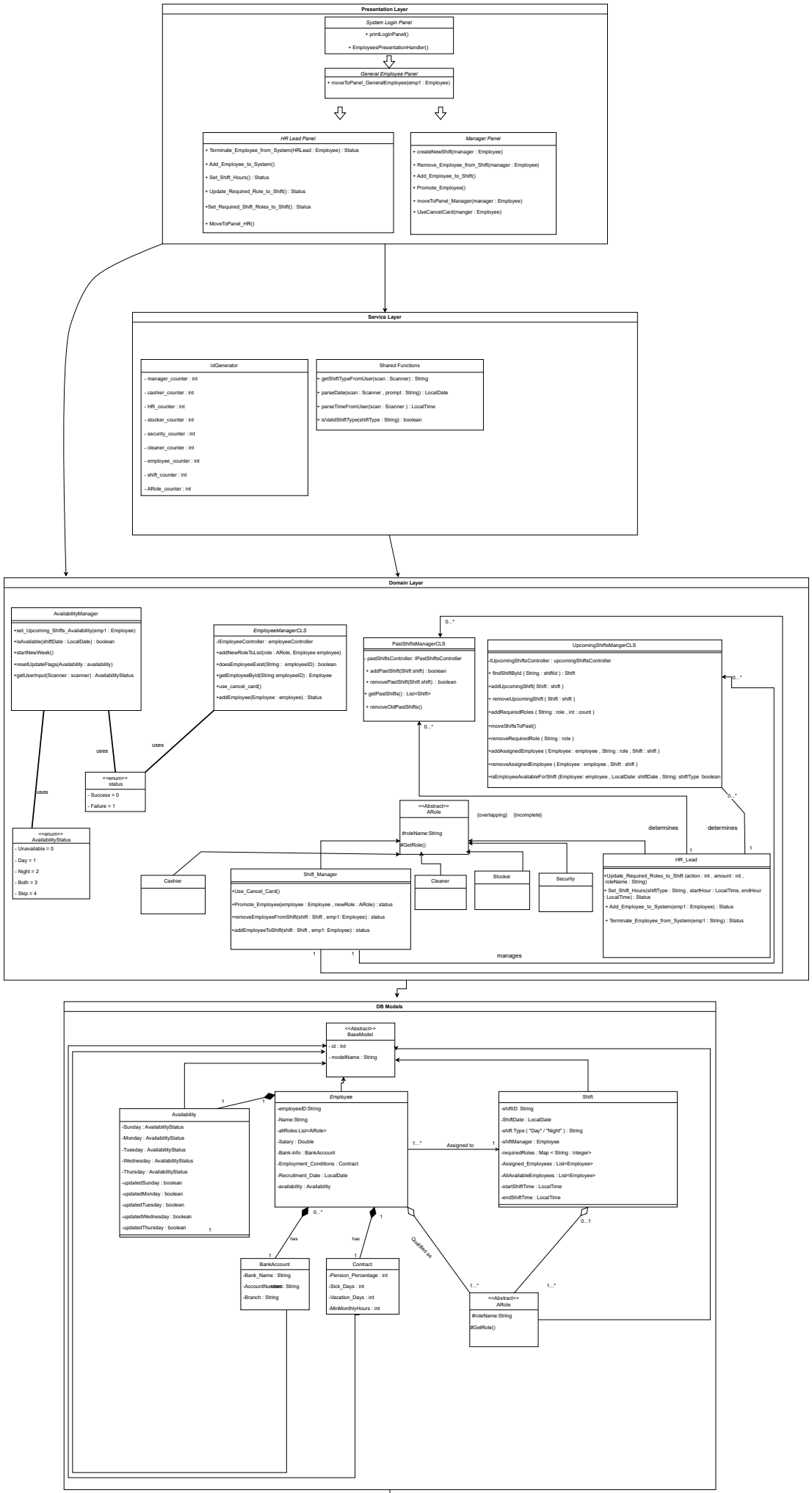




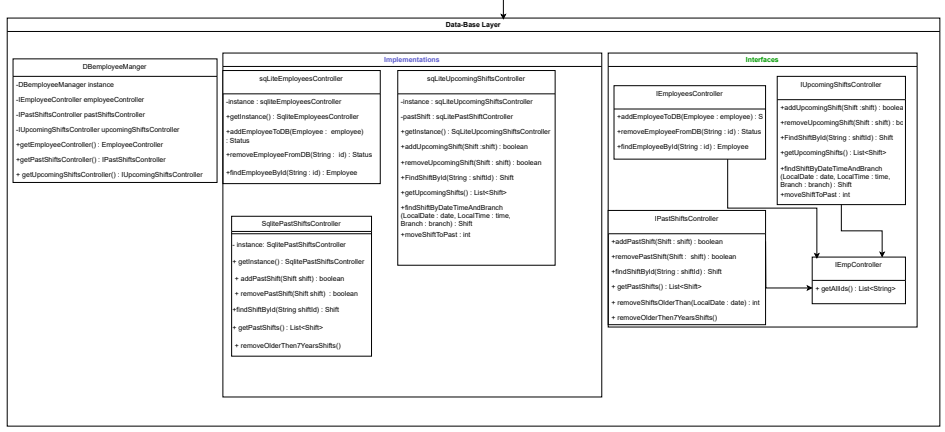
Linkage connection




abl1 : Availability
-Sunday : 2
-Monday : 3
-Tuesday : 1
-Wednesday : 0
-Thursday : 0
-updatedSunday : False
-updatedMonday : False
-updatedTuesday : True
-updatedWednesday : True
-updatedThursday : True
abl2 : Availability
-Sunday : 1
-Monday : 1
-Tuesday : 3
-Wednesday : 0
-Thursday : 2
-updatedSunday : True
-updatedMonday : True
-updatedTuesday : True
-updatedWednesday : True
-updatedThursday : True
emp2 : Employee
-employeeID: "202408247"
-Name: "Yael Dur"
-Roles: {Cashier, Cleaner , HR Lead }
-Salary : 10.00
-Bank-_nfo : Ox400060 (pointer to BankAccount Objec
-Employment Conditions : Ox468000 (pointer)
-Recruitment_Date : 11/09/2024
-availability : Ox400400 (pointer to Availability Object)
sft2 : Shift
-shiftID : "0160"
-Date : 16/04/2025
-Assigned Employees : {Yael , Mohamad , Alon , Eitan }
-AllAvailableEmployees : {Sali , Ben , Shay , Irit }
-Shift Type ("Day" / "Night") : "Day"
-Start_Shift_Time : 08:00
-Finish_Shift_Time : 15:00
-ShiftManager : Denis
-RequiredRoles : {Stocker: 1 , Cashier : 4, Shift_Manger: 1}
Assigned to
Manages
emp1 : Employee
-employeeID: "202678247"
-Name: "Ron Levi"
-Roles: {Cashier }
-Salary : 100.00
-Bank-info : Ox409060 (pointer to BankAccount Object
-Employment Conditions : Ox468000 (pointer)
-Recruitment Date : 03/07/2024
-availability : Ox406400 (pointer to Availability Object)
has
has
bank2 : BankAccount
-Bank_Name : "Leumi"
-AccountNumber : "074651"
-Branch : "601"
cont2 : Contract
-Pension_Percentage : 6
-Sick_Days : 28
-Vacation_Days : 14
-MinMonthlyHours : 100
Qualified as
determines
Assigned to
Qualified as
has
has
bank1 : BankAccount
-Bank_Name : "Yahav"
-AccountNumber : "174351"
-Branch : "71"
cont1: Contract
-Pension_Percentage : 8
-Sick_Days : 14
-Vacation_Days : 10
-MinMonthlyHours : 80
sft1 : Shift
-shiftID: "7660"
-Date : 18/04/2024
-Assigned Employees : {Yael , Ron , Eitan }
-AllAvailableEmployees : {Denis , Tom , Shay , Irit }
-Shift Type ("Day" / "Night") : "Night"
-Start_Shift_Time : 15:00
-Finish_Shift_Time : 22:00
-ShiftManager : Tali
-RequiredRoles : {Cashir: 2 , Cleaner: 4, Shift_Manger: 1}
<<Abstract>>
ARole
{overlapping}
{incomplete}
Cashier
Shift_Manger
Cleaner
Stocker
Security
HR_Lead
Manages
determines

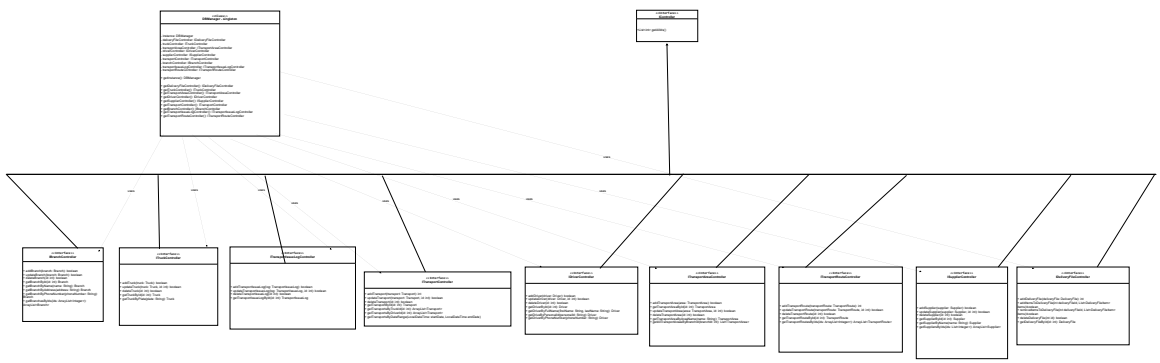

Presentation Layer
System Login Panel
+ printLoginPanel()
+ EmployeesPresentationHandler()
General Employee Panel
+ moveToPanel_GeneralEmployee(empt : Employee)
HR Lead Panel
+ Terminate_Employee_from_System(HRLead : Employee) : Status
+ Add_Employee_to_System()
+ Set_Shift_Hours() : Status
+ Update_Required_Role_to_Shift() : Status
+Set_Required_Shift_Roles_to_Shift() : Status
+ MoveToPanel_HR()
Manager Panel
+ createNewShift(manager : Employee)
+ Remove_Employee_from_Shift(manager : Employee)
+ Add_Employee_to_Shift()
+ Promote_Employee()
+ moveToPanel_Manager(manager : Employee)
+ UseCancelCard(manger : Employee)
Service Layer
IdGenerator
- manager_counter : int
- cashier_counter : int
- HR_counter : int
- stocker_counter : int
- security_counter : int
- cleaner_counter : int
- employee_counter : int
- shift_counter : int
- ARole_counter : int
Shared Functions
+ getShiftTypeFromUser(scan : Scanner) : String
+ parseDate(scan : Scanner , prompt : String) : LocalDate
+ parseTimeFromUser(scan : Scanner) : LocalTime
+ isValidShiftType(shiftType : String) : boolean
Domain Layer
AvailabilityManager
+set_Upcoming_Shifts_Availability(emp1 : Employee)
+isAvailable(shiftDate : LocalDate) : boolean
+startNewWeek()
+resetUpdateFlags(Availability : availability)
+getUserInput(Scanner : scanner) : AvailabilityStatus
EmployeeManagerCLS
-IEmployeeController : employeeController
+addNewRoleToList(role : ARole, Employee employee)
+doesEmployeeExist(String : employeeID) : boolean
+getEmployeeById(String employeeID) : Employee
+use_cancel_card()
+addEmployee(Employee : employee) : Status
PastShiftsManagerCLS
- pastShiftsController: IPastShiftsController
+ addPastShift(Shift shift) : boolean
+ removePastShift(Shift shift) : boolean
+ getPastShifts() : List<Shift>
+ removeOldPastShifts()
UpcomingShiftsMangerCLS
-IUpcomingShiftsController : upcomingShiftsController
+ findShiftById (String : shiftId) : Shift
+addUpcomingShift(Shift : shift)
+ removeUpcomingShift (Shift : shift)
+addRequiredRoles (String : role , Int : count)
+moveShiftsToPast()
+removeRequiredRole (String : role)
+addAssignedEmployee (Employee : employee , String : role , Shift : shift)
+removeAssignedEmployee (Employee : employee , Shift : shift)
+isEmployeeAvailableForShift (Employee: employee , LocalDate: shiftDate , String: shiftType :boolean
<<enum>>
status
- Success = 0
- Failure = 1
<<enum>>
AvailabilityStatus
- Unavailable = 0
- Day = 1
- Night = 2
- Both = 3
- Skip = 4
<<Abstract>>
ARole
#roleName:String
#GetRole()
{overlapping} {incomplete}
Cashier
Shift_Manager
+Use_Cancel_Card()
+Promote_Employee(employee : Employee , newRole : ARole) : status
+removeEmployeeFromShift(shift : Shift , emp1: Employee) : status
+addEmployeeToShift(shift : Shift , emp1: Employee) : status
Cleaner
Stocker
Security
HR_Lead
+Update_Required_Roles_to_Shift (action : int , amount : int , roleName : String)
+ Set_Shift_Hours(shiftType : String , startHour : LocalTime, endHour : LocalTime) : Status
+ Add_Employee_to_System(emp1 : Employee) : Status
+ Terminate_Employee_from_System(emp1 : String) : Status
uses
uses
uses
determines
determines
manages
0...*
0...*
0...*
1
1
1
1

DB Models
<<Abstract>>
BaseModel
- id : Int
- modelName : String
Employee
-employeeID:String
-Name:String
-allRoles:List<ARole>
-Salary : Double
-Bank-info : BankAccount
-Employment_Conditions : Contract
-Recruitment_Date : LocalDate
-availability : Availability
Availability
-Sunday : AvailabilityStatus
-Monday : AvailabilityStatus
-Tuesday : AvailabilityStatus
-Wednesday : AvailabilityStatus
-Thursday : AvailabilityStatus
-updatedSunday : boolean
-updatedMonday : boolean
-updatedTuesday : boolean
-updatedWednesday : boolean
-updatedThursday : boolean
Shift
-shiftID: String
-ShiftDate : LocalDate
-shift Type ("Day" / "Night") : String
-shiftManager : Employee
-requiredRoles : Map < String : Integer>
-Assigned_Employees : List<Employee>
-AllAvailableEmployees : List<Employee>
-startShiftTime : LocalTime
-endShiftTime : LocalTime
Assigned to
has
has
Qualified as
BankAccount
-Bank_Name : String
-AccountNumber: String
-Branch : String
Contract
-Pension_Percentage : int
-Sick_Days : int
-Vacation_Days : int
-MinMonthlyHours : int
<<Abstract>>
ARole
#roleName:String
#GetRole()
1..*
0...1
1...*
1...*
1
1
1
1
0...*
1
1
1


Data-Base Layer
DBemployeeManger
-DBemployeeManager instance
-IEmployeeController employeeController
-IPastShiftsController pastShiftsController
-IUpcomingShiftsController upcomingShiftsController
+getEmployeeeController() : EmployeeController
+getPastShiftsController() : IPastShiftsController
+ getUpcomingShiftsController() : IUpcomingShiftsController
Implementations
sqLiteEmployeesController
-instance : sqliteEmployeesController
+getInstance() : SqliteEmployeesController
+addEmployeeToDB(Employee : employee) : Status
+removeEmployeeFromDB(String : id) : Status
+findEmployeeById(String : id) : Employee
sqLiteUpcomingShiftsController
-instance : sqLiteUpcomingShiftsController
-pastShift : sqLitePastShiftController
+getInstance() : SqLiteUpcomingShiftsController
+addUpcomingShift(Shift :shift) : boolean
+removeUpcomingShift(Shift : shift) : boolean
+FindShiftById(String : shiftId) : Shift
+getUpcomingShifts() : List<Shift>
+findShiftByDateTimeAndBranch (LocalDate : date, LocalTime : time, Branch : branch) : Shift
+moveShiftToPast : int
SqlitePastShiftsController
- instance: SqlitePastShiftsController
+ getInstance() : SqlitePastShiftsController
+ addPastShift(Shift shift) : boolean
+ removePastShift(Shift shift) : boolean
+findShiftById(String shiftId) : Shift
+ getPastShifts() : List<Shift>
+ removeOlderThen7YearsShifts()
Interfaces
IEmployeesController
+addEmployeeToDB(Employee : employee) : S
+removeEmployeeFromDB(String : id) : Status
+findEmployeeById(String : id) : Employee
IUpcomingShiftsController
+addUpcomingShift(Shift : shift) : boolea
+removeUpcomingShift(Shift : shift) : bo
+FindShiftById(String : shiftId) : Shift
+getUpcomingShifts() : List<Shift>
+findShiftByDateTimeAndBranch (LocalDate : date, LocalTime : time, Branch : branch) : Shift
+moveShiftToPast : int
IPastShiftsController
+addPastShift(Shift : shift) : boolean
+removePastShift(Shift : shift) : boolean
+findShiftById(String : shiftId) : Shift
+ getPastShifts() : List<Shift>
+ removeShiftsOlderThan(LocalDate : date) : int
+ removeOlderThen7YearsShifts()
IEmpController
+ getAllIds() : List<String>

DB Layer

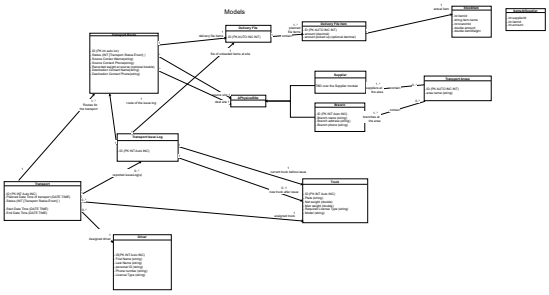
Controllers



Enums



Models

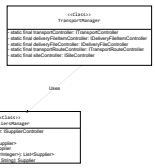


Domain Layer

Logical Manager Classes

Enums

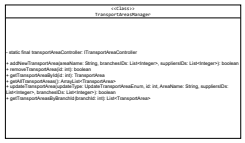


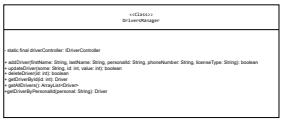




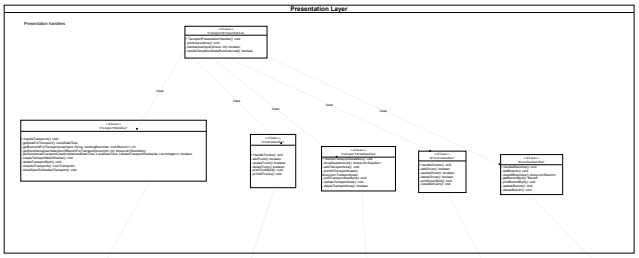
Uses

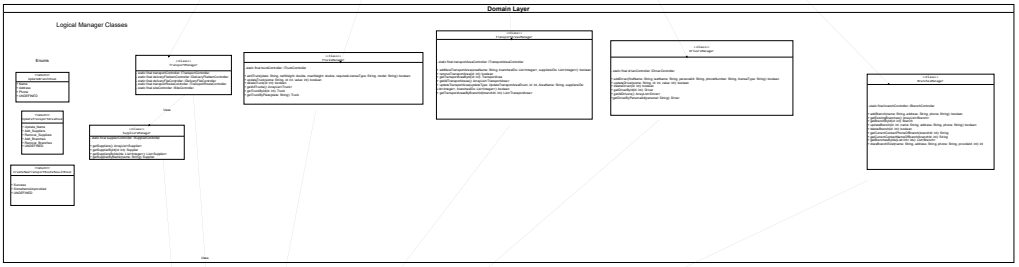


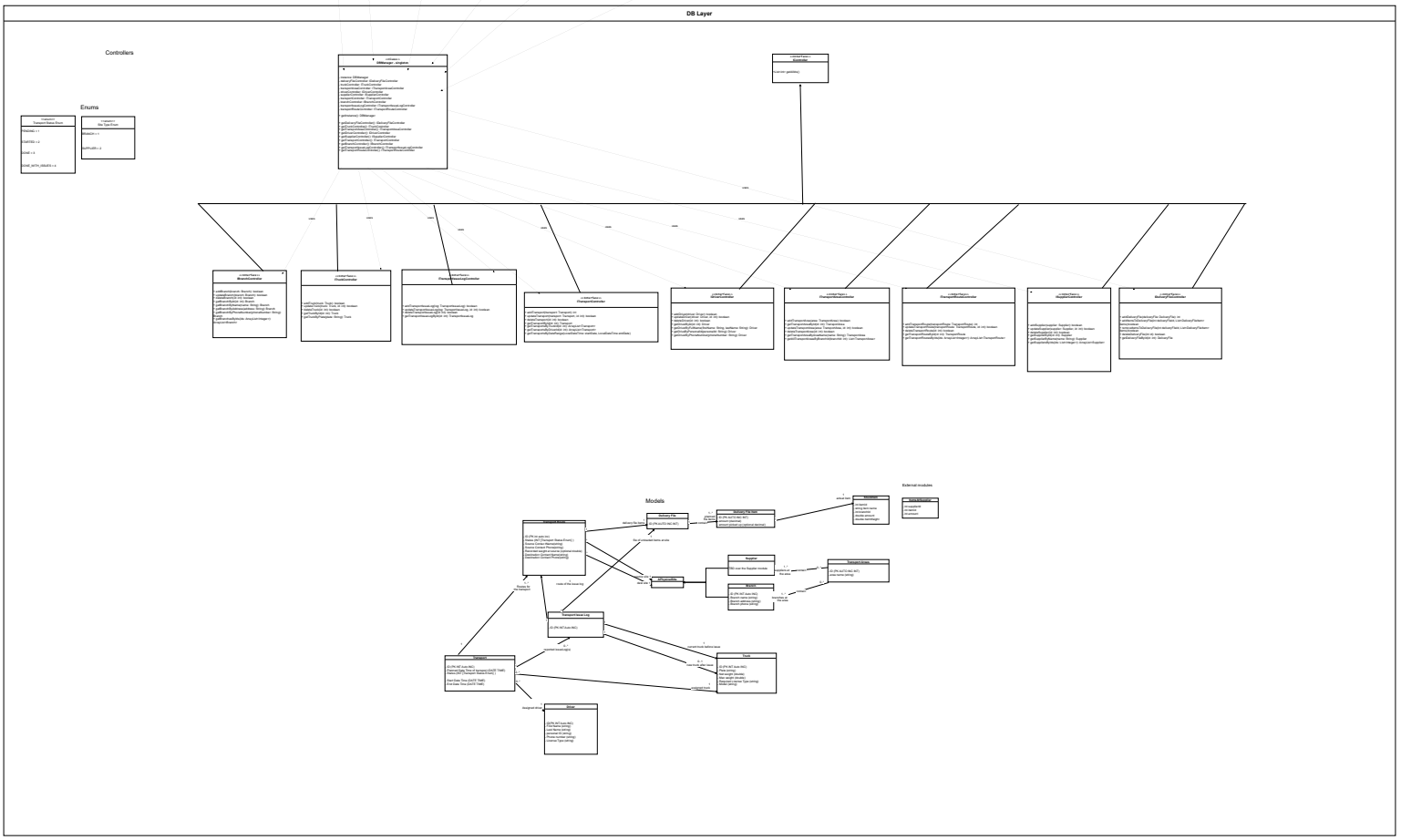






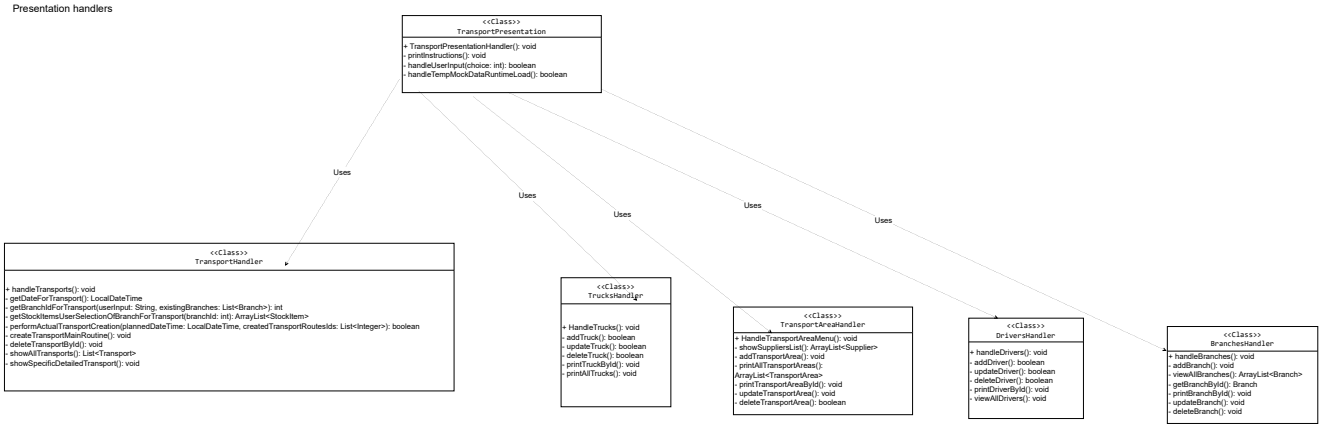

Presentation Layer
Presentation handlers


Domain Layer
Logical Manager Classes
Enums


DB Layer
Controllers
Enums
Models
External modules

Presentation Layer

Presentation handlers



ID	Module	Functional/ Non-Functional	Description	Priority	Risk	Status
1.	Workers	F	The system **MUST** support registration of new employees	**MH**	High	Done
2.	Workers	F	The system **MUST** support the termination of an employee	**MH**	High	Done
3.	Workers	F	The system **MUST** contain employees's Name,ID,Bank-info, Salary, employement conditions ,reqruitment date and all role capabilties	**MH**	High	Done
4.	Workers	F	The system **MUST** consider each of employee's schedule restrictions before assigning them into shifts (days + hours)	**MH**	High	Done
5.	Workers	F	The system **MUST** store all employee shift schedule restrictions and preferencesfor scheduling decisions	**MH**	Low	Done
6.	Workers	F	The system **MUST** support to receive employees shifts restrictions until every Thursday.	**MH**	Low	Done
7.	Workers	NF	The system **MUST** contain shifts history for 7 years	**MH**	Low	Done
8.	Workers	F	The system **MUST** allow only users with the role 'Shift Manager' to assign personnel to shifts via the shift management interface.	**MH**	High	Done

9.	Workers	F	The system **MUST** give permission to the HR Lead to update upcoming shift hours	**MH**	Low	Done
10.	Workers	F	The system **SHOULD** give permission to the Shift Manger to update the upcoming shift's emploees even after publishing the upcoming shift	**NTH**	High	Done
11.	Workers	F	The system **SHOULD** warn the manager for committing changes that overstep the boundaries set by the HR lead.	**NTH**	Low	Done
12.	Workers	F	The system **MUST** let only to each Shift Manger to cancel items	**MH**	Low	Done
13.	Workers	NF	The system **MUST** prohibit inserting unqualified personnel into roles for which they lack the required training	**MH**	Low	Done
14.	Workers	F	The system **MUST** confirm that each shift will have ONLY one Shift Manger and the designated ammount of perssonel by the HR	**MH**	Low	Done
15.	Workers	NF	The system **MUST** distinguish between different role logins by ID	**MH**	Low	Done
16.	Workers	F	The system **SHOULD** allow real-time notifications when an employee is assigned or removed from a shift	**NTH**	High	Wont be implmented

17.	Workers	F	The system **SHOULD** allow employees to request shift swaps with other employees through the platform	**NTH**	High	Wont be implmented
18.	Workers	F	The system **SHOULD** send automatic reminders to employees about upcoming shifts and changes	**NTH**	Low	Wont be implmented
19.	Workers	NF	The system **SHOULD** support multilingual UI (Hebrew and English) for ease of use by diverse staff	**NTH**	High	Wont be implmented
20.	Workers	F	The system **SHOULD** allow exporting shift reports/history to PDF	**NTH**	Low	Wont be implmented
21.	Workers & Transportaion	NF	The system **MUST** support scalability to allow the addition of multiple branches.	**MH**	Low	TBD
22.	Workers & Transportaion	F	The system **MUST** support sqLite queries for dataBase usage	**MH**	High	TBD
23.	Workers & Transportaion	F	The system **MUST** ensure ensure drivers assigned to a transport route have a matching shift in the Employees system.	**MH**	High	TBD
24.	Workers & Transportaion	F	The system **MUST** Use consistent Employee.id across both modules (when Driver is an Employee) to avoid desynchronization or duplicate references.	**MH**	Low	TBD

25.	Workers & Transportaion	F	The system **MUST** Validate that any employee assigned to a transportation route has the "Driver" role	**MH**	Low	TBD
26.	Workers & Transportaion	F	Upon creating a new transport route, The system **SHOULD** auto suggest or pre-fill a shift for the driver in the Employees DB, including expected time and branch.	**NTH**	High	TBD
27.	Workers & Transportaion	F	The system **SHOULD** log or alert if a transport driver is double-booked or has overlapping shifts/routes	**NTH**	High	TBD

ID	Module	Functional / Non-Functional	Description	Priority	Risk	Status	
1	Transport	NF	The System must follow this format for Delivery File Item: - ID - Item ID(FK) - amount - amout picked up	MH	High	DONE	
2	Transport	NF	The System must follow this format for Delivery File: - ID - list of Delivery File Item(s) IDs(FKs)	MH	High	DONE	
3	Transport	NF	The System must follow this format for Truck: - ID(Auto INC) - Plate - Neto weight - max weight - Required License Type - Model	MH	High	DONE	

4	Transport	NF	The System must follow this format for Driver: - ID(Auto INC) - First Name - Last Name - personal ID - Phone number - License Type	MH	High	DONE	
5	Transport	NF	The System must follow this format for Branch: - ID - Branch name - Branch address - Branch phone	MH	High	DONE	
6	Transport	NF	The System must follow this format for APhysicalSite: - ID - Site name - Site address - Site phone Branch inheritance from that	MH	High	DONE	

7	Transport	NF	The System must follow this format for Transport Issue Log: - Current Truck ID - New Truck ID - Unloaded items at Site ID - Unloaded Items Delivery File ID - Transport route ID	MH	High	DONE	
8	Transport	NF	For any Transport in the System, a Transport should have: - Transport ID - planned datetime of transport - status (pending, started, done, done with issues) - Truck ID (FK to link to the actual truck) - Driver ID (FK to link to the actual driver) - source - List of APhysicalSite(s) ID(s) (FKs) - destination - List of APhysicalSite(s) ID(s) (FKs) - List of Transport Issue Log ID(s) (FK) - List of TransportRoutes ID(s) (Fk) - actual start datetime - actual end datetime	MH	High	DONE	

9	Transport	NF	The System must follow this format for TransportAreas: - ID -area name (string)	MH	High	DONE	
10	Transport	NF	For any TransportRoute in the System, a TransportRoute should have: - TransportRoute ID - Status(PENDING,STARTED,END..) -Source site ID (FK) -Destination site ID (FK) - Delivery File ID (FK)	MH	High	DONE	
11	Transport	NF	The System must send a Transport for dependent Suppliers only.	MH	High	DONE	

12	Transport	NF	In a given Transport the System must know from which suppliers to which branches the supplies should be distribued, via TransportRoute	MH	High	DONE	
13	Transport	F	The system must allow to check for requested items at supplier - if they exist - in which suppliers	NTH	Low	DONE	
14	Transport	F	The System must allow an admin to assign a user to the role of Transport Manager	NTH	Low	TBD	
14	Transport	F	The System must allow to define a user as admin	NTH	High	TBD	
15	Transport	F	The System must allow to define a user as transport manager	NTH	High	TBD	
16	Transport	F	The System must allow a transport manager/user to create a future transport	MH	High	DONE (SUPPORT ONLY USER)	
17	Transport	F	The System must allow a transport manager/user to update a transport	MH	High	DONE (SUPPORT ONLY USER)	
18	Transport	F	The System must allow a transport manager/user to delete a transport	MH	High	DONE (SUPPORT ONLY USER)	
19	Transport	F	The System must allow a transport manager/user a transport manager to create a driver	MH	High	DONE (SUPPORT ONLY USER)	

20	Transport	F	The System must allow a transport manager/user to edit a driver	MH	High	DONE (SUPPORT ONLY USER)	
21	Transport	F	The System must allow a transport manager/user to delete a driver	MH	High	DONE (SUPPORT ONLY USER)	
22	Transport	F	The System must allow a transport manager/user to create a truck	MH	High	DONE (SUPPORT ONLY USER)	
23	Transport	F	The System must allow a transport manager/user to edit a truck	MH	High	DONE (SUPPORT ONLY USER)	
24	Transport	F	The System must allow a transport manager/user to delete a truck	MH	High	DONE (SUPPORT ONLY USER)	
25	Transport	F	The System must allow a admin/user to create a branch	MH	High	DONE (SUPPORT ONLY USER)	
26	Transport	F	The System must allow a admin/user to edit a branch	MH	High	DONE (SUPPORT ONLY USER)	
27	Transport	F	The System must allow a admin/user to delete a branch	MH	High	DONE (SUPPORT ONLY USER)	
29	Transport	F	The System must allow a admin/user to create a TransportArea	MH	High	TBD	
30	Transport	F	The System must allow a admin/user to edit a TransportArea	MH	High	TBD	

31	Transport	F	The System must allow a admin/user to delete a TransportArea	MH	High	TBD	
32	Transport	F	During a given Transport - if a requested item(s) is missing from the relevant supplier - The System must allow the truck to be re-routed to a different supplier(s) to receive the missing item(s).	NTH	High	TBD	
33	Transport	F	When a Transport starts - the System must change the status of the Transport(status + datetime of the start), in all the Transport routes that connected to it.	NTH	High	DONE	
34	Transport	F	The System must alert in case that the initial truck weight is above the maximum truck weight	MH	Low	DONE	
35	Transport	F	Upon maximum truck weight overflow - the system will ask the TransportManager/user which strategy to use: - cancel or change one of the destinations - change truck - unload certain items from the current truck (Basically give the option to edit the entire transport)	MH	High	DONE (SUPPORT ONLY USER)	

36	Transport	F	The System can not allow a driver without a sufficent drivers license type to drive any truck	MH	High	DONE	
37	Transport	F	The System have to show the TransportArea by ID	MH	High	DONE	
38	Transport	F	The System have to show all the TransportAreas in the System	MH	High	DONE	
39	Transport	F	The System have to have show all the Transport in the System	MH	High	DONE	
40	Transport	F	The System should have show all the Transport by ID in the System	MH	High	DONE	
41	Transport	F	The System have to Print transport drives for driver - by Transport ID	NTH	High	DONE	
42	Transport	F	The System have to allow to Visit source of Transport Transport's route	NTH	High	DONE	
43	Transport	F	The System have to allow to destination source of Transport Transport's route	NTH	High	DONE	
44	Transport	F	The System have to show truck by its ID	MH	High	DONE	
45	Transport	F	The System have to show all the trucks in the System	MH	High	DONE	
46	Transport	F	The System have to show drivers by its ID	MH	High	DONE	
47	Transport	F	The System have to show all the drivers in the System	MH	High	DONE	
48	Transport	F	The System have to show branch by its ID	MH	High	DONE	

49	Transport	F	The System have to show all the branch in the System	MH	High	DONE	
50	Transport	F	The System have to show gui for handle all the models	MH	High	DONE	
51	Transport	NF	The System have to load/unload by the chioce of the user mock values (only for this version)	MH	High	DONE	
52	Transport	NF	The system must check if there is at least one available Stocker by date and time to start the Transport	MH	High	TBD	
53	Transport	NF	The system must check if there is at least one available Driver by time and date to start the Transport	MH	High	TBD	
54	Transport	NF	The system must synchornize with Employ system that drivers will be one to one with employ table in Employ system	MH	High	TBD	

ID	Module	Func/Non-Func	Description	Priority	Risk	Status
1	Stock	F	The system MUST give a notification before an item runs out of stock	MH	HIGH	TBD
2	Stock	F	The system MUST provide the option to choose the minimum quantity of the product before a notification is sent.	MH	LOW	DONE
3	Stock	NF	The system MUST follow this format for each product info: product ID amount, where, manufacturer, amount on shelves, amount in the warehouse.	MH	LOW	DONE
4	Stock	F	The system MUST be able to generate stock status for all the products in the store	MH	HIGH	DONE
5	Stock	NF	the system MUST follow this format for category: product ID, purchase price, amount in warehouse, amount in store shelves, sells price.	MH	LOW	DONE
6	Stock	NF	the System MUST follow the format for damaged products: BatchID, product ID, LocationType(warehouse,store shelves).	MH	LOW	DONE

			amount.			
7	Stock	F	The system MUST be able to make a report based on a category or categories	MH	HIGH	DONE
8	Stock	F	The system MUST be able to make a report of all defective items	MH	HIGH	DONE
9	Stock	F	The system MUST have the option update the amount of the products	MH	Low	DONE
10	Stock	F	The System MUST have the option to add a new product to the stock	MH	HIGH	DONE
11	Stock	F	The System MUST have the option to put to sale a category	MH	HIGH	DONE
12	Stock	F	The System MUST return all the products of a same category	MH	HIGH	TBD
13	Stock	F	the System MUST generate for every product a unique id	MH	llow	DONE
14	Stock	F	the System MUST generate for each category a unique id.	MH	low	DONE
15	Stock	F	the system MUST sort the products into three category each subcategory of the other	MH	HIGH	DONE
16	Stock	F	the System MUST to have the option to apply a discount on product or category(not both)	MH	HIGH	DONE
17	Stock	F	the system MUST track the product expiration date	MH	HIGH	DONE
18	Stock	F	the SHOULD support of moving product from the warehouse to the store shelve	NTH	LOW	DONE
19	Suppliers	F	the system must support adding new suppliers and saving their information in the format:			Done

			name , address , phone , number (unique and not null) , payment method , payment time , bank details and contact people which is a list of <name , phone number>			
20	Suppliers	F	the system will allow updating the details of existing suppliers in the system (the details are mentioned in 19)			Done
21	Suppliers	F	the system will allow deleting suppliers from the system			Done
22	Suppliers	F	the system will support creating orders from suppliers in the system , an order will contain the following information: supplier information , address , date , phone , status, list of the products in the order			Done
23	Suppliers	F	the system will allow updating the details of existing orders if they are in a non delivered status			Yet to start
24	Suppliers	F	the system will allow deletion of orders			Done
25	Suppliers	F	the system will keep track of status of orders and allow the user to change the status of the orders as he likes			Done

26	Suppliers	F	the system will support the functionality of showing orders by their status and in general all the orders as the user likes , tin this functionality the system will show the orders in a shorter form: supplier info, address , date , phone , id, status			Done
27	Suppliers	F	the system will allow showing the full details of an order (including the list of items,quantity as well as price descriptions) by id			Done
28	Suppliers	F	the system will support showing all the suppliers in the system in a shortened form as follows: name , address , supplier's number , payment method , payment time , bank details			Done
29	Suppliers	F	the system will allow showing the full details of a supplier by his id , the details as mentioned in the requirement 19			Done
30	Suppliers	F	the system should support searching suppliers by different categories : name,number			Done
31	Suppliers	F	the system should support searching orders by different categories : by id			Done

32	Suppliers	F	the system should make the range of categories which by the users can search orders and suppliers as big as possible (by every field of supplier/order)			Yet to start
33	Suppliers	F	the system should support combination of search categories , for example , the users can search suppliers by names and address			Yet to start
34	Suppliers	F	the system must support assigning contracts to suppliers in the system , a contract should have the following information : supplier information , list of contract terms contract term contains a product's information , price and an amount specifier amount specifier contains information regarding discount per quantity , so minimal quantity to receive the discount , base discount percentage			Done
35	Suppliers	F	the system should support adding amount specifiers to contracts , an amount specifier is explained in 34			Done
36	Suppliers	F	the system should support editing existing amount specifiers which are			Yet to start

			already attached to contract terms , explained in 34			
37	Suppliers	F	the system should support removal of existing amount specifiers which are already attached to contract terms , explained in 34			Yet to start
38	Suppliers	F	the system should allow updating a contract details , as explained in 34			Yet to start
39	Suppliers	F	the system will allow deletion of existing contracts as explained in 34			Yet to start
40	Suppliers	F	the system must allow the user to view history of purchases as well as meta data about the overall history : total sum , number of products purchased			Done
41	Suppliers	NF	the system must respond within 10 seconds for each user interaction with probability of 99%			Done
42	Suppliers	NF	the system should be simple to use with clear user interfaces and clear options			Done
43	Suppliers	NF	the system should save important information : orders,suppliers,contracts			Done

44	Suppliers	NF	the system must save it’s state from previous interactions			Yet to start
45	Suppliers	NF	the system should be well-oriented to allow users to reach any wanted page in very few interactions (2 - 4) with clear paths			Done
46	Suppliers& Stock		the system MUST have the option to add a new periodic order which have information as normal orders as explained in 22 , as well as delivery day			TBD
47	Suppliers& Stock		the system must allow the user to trigger automatic creation of orders which should make every product in the stock about it’s own critical level .			TBD
48	Suppliers& Stock		the system will support updating periodic orders however it will limit the time for updating to one day before the date of the delivery of the order .			TBD
49	Suppliers& Stock		the system will support showing a list of all products which are under their critical level , the products will be shown in a shorten form : product id , current quantity and critical level			TBD

דרישות פתוחות:

#	Topic	Issue
1	מחיקה של ספקים	אפשר לעדכן או למחוק ספקים מהמערכת ?
2	מספק לא קיים	אפשר הזמנה מספק שלא קיים במערכת ? אם כן אז איך תתנהג ת במצב כזה ?
3	הזמנות	אפשר עדכון הזמנות אחרי שכבר נוצרו ? אם כן אז האם יש תנאים זו ?
4	הזמנות	מערכת צריכה לעקוב אחרי סטטוס של כל הזמנה ? אם כן אז האם למשתמש לעדכן אותה או רק על ידי המערכת או הרשאות מיוחדות ?
5	המערכת	צופה מהמערכת להגיב "במהירות" ? אם כן אז נא להגדיר סף
6	ויות בהסכמים	חייב צירוף כתבי כמויות רק בעת הכנסת הסכם למערכת ? או לנהל אותם לגמרי ?
7	ספקים	' המערכת לאפשר הצגה של כל הספקים ? אם כן אז איך בדיוק ?
8	הזמנות	' המערכת לתמוך בהצגת הזמנות ? אם כן אז איך בדיוק ?
9	לספקים	' המערכת לאפשר חיפושים לפי קטגוריות עבור ספקים ?
10	הזמנות	' המערכת לאפשר חיפוש הזמנות ומיונם/סינונם לפי קטגוריות ?
11	הסכמים	' המערכת לאפשר חיפוש הסכמים לפי קטגוריות ובנוסף לאפשר חיפוש ק/פריט מסוים ?

Does the system need to provide the option to add new stock? ans:the system should have the option to add new products.	add new stock	12
Does the management want the system to calculate the sale or only to give the information about a specific product?	In the store, various sales can be managed for different products.	13
the report should present the total amount of products under same category or for each product under the category show the total amount? ans:the report presents the total amount in each category.	category report	14
witch discount applies if you have two discount on a product? ans: the discount with the most up to date start date will be preferable	Discount	15
Does the program need to keep a list of all the discounts? ans: no	Discount	16
Does the program need to search products by their name or ID? ans: ID	Products search	17
In the category stock report should the report should the stock per category and count each item once or for each category with no repeating items? ans: no repeating items	Category report	18
If a product is damaged should it be counted as an existing item to know if we need to buy more from the supplier?	Item counter	19

ans: should not be counted		
If an item is low in stock how much do we need to order? ans: order an item to the low stock quantity	order stock	20

#	Topic	Issue	
1.	Employment Contract	What are the employment conditions?	Sick Days, Vacation Days, Pension, Min Working hours per month,
2.	Salary	Global salary or Hourly rate ?	Hourly
3.	Numebr of Store	How many stores are opening	For now only One.
4.	SuperMarket Roles	What Roles are there in the SuperMarket?	Cashier , Manger , Stocker , Cleaner , Security , HR Lead
5.	Permissions Hierarchy	Who can Accept / Terminate employees?	Only HR Lead

<u>שינויים בטבלאות דרישות מודל העובדים מאז חלק א:</u>

1) בעקבות ההערות מחלק א של העבודה הבנו שדרישות 11, 16 היו מנוסחות בצורה לא מדויקת ולכן דייקנו את הניסוח.
   בנוסף הבנו ששתי הדרישות הללו אמורות להיות פונקציונליות ולא NF כי הדרישות הללו מתאימו יותר להגדרה של דרישה פונקציונלית מאשר דרישה לא פונקציונלית.

2) דרישה מס 8 הייתה מנוסחת בצורה לא טובה כי לא דייקה את הדרישה שהתכוונו אליה לכן כעת ניסחנו את הדרישה שתתאים למימוש אנחנו מתכוונים לעשות בקוד שלנו.

3) הוספנו את דרישות 21-27 על מנת לאפשר אינטגרציה (שילוב) חלקה עם מודל ההובלות.

<u>שינויים בטבלת ספקים ומחסן:</u>

1)צירפנו את הטבלה של שאר המודלים(עובדים והובלות)

2) שינינו את טבלת הספקים כך שהיא תתאים לפורמט של שאר הטבלאות(באנגלית)

3)הוספנו את 46-49 בשביל שיהיה איחוד וקישור בין המודלים מחסן וספקים

<u>הגדרות עבור מודל העובדים:</u>

מנהל כוח אדם (HR):
עובד בעל הרשאות ניהול גבוהות, אחראי על הוספה והסרה של עובדים מהמערכת, קביעת שעות משמרת, הגדרת הדרישות לסוגי התפקידים במשמרת והכמות מכל אחד מהם.

מנהל משמרת:
עובד עם הרשאות ניהול משמרות, יכול ליצור משמרות, לשבץ ולהסיר עובדים מהן, לקדם עובדים לתפקידים נוספים ולבצע פעולות ניהוליות הקשורות למשמרות בלבד.

עובד כללי :
משתמש במערכת ללא הרשאות ניהול, יכול לעדכן זמינות למשמרות בשבוע הקרוב, לצפות בפרטיו האישיים ולבצע פעולות בסיסיות הקשורות לעבודתו בלבד.
כרגע העובדים הכללים במערכת הם: קופאי\ת, מחסנאי\ת, מנקה ונהג.

זמינות :
היכולת של עובד לבחור או לעדכן באילו ימים או שעות הוא פנוי לעבוד, לצורך שיבוץ למשמרות.
עידכון הזמינות מתאפשר לעשות רק עד שבוע קדימה.

חוזה :
המידע הבסיסי שמגדיר את תפקידו של העובד בארגון, כמות ימי החופשות, אחוז הפנסייה, כמות שעות עבודה מינימלית בחודש ועוד.

תפקיד :
במערכת שלנו קיימים 7 תפקידים- קופאי, מנקה, מחסנאי, מנהל משמרת, מנהל כוח אדם, נהג ומאבטח.

<u>הגדרות עבור מודל הובלות:</u>

מנהל מערכת - אחראי על הגדרת מנהל הובלה (Administer)

מנהל הובלה - אחראי על יצור של הובלות חדשות, מחיקה של הובלה קיימת, עדכון של הובלה קיימת

בנוסף אחראי על סידור נהגים ומשאיות לפי כל הובלה

מסלול הובלה - מסלול הובלה מספק אחד אל יעד אחד על ידי נהג יחיד ומשאית יחידה. בדיקה בכל מסלול אם המשאית עומדת במשקל ואם לא חסרים פריטים אשר אמורים להגיע ליעד. בנוסף, בדיקה אם קיים מחסנאי זמין ביעד לסדר את הסחורה.

הובלה - אוסף של מסלולי הובלה כאשר מצוות לכל המסלולים נהג אחד ומשאית אחת, בדיקה שכל התנאים מתקיימים וניתן להוציא הובלה לפועל.

נהג - עובד אשר הוגדר כנהג לפי רישיון מתאים וכדומה

משאית - כלי רכב שהוגדר לפי משקל, סוג רישיון וכדומה.

<u>הגדרות מודל מחסן:</u>

product - מוצר אשר נימצא במחסן בעל מספר מזהה,שם, יצרן,וכמה יש ממנו בכללי בתוך החנות.

productBatch- כמות מסויימת של מוצר מסויים מאופיין על ידי מס מזהה באצ', מוצר, כמות מאותו המוצר והיא הוא נימצא וגם תאריך תפוגה שלו ואם המוצר פגום או לא. כלומר למוצר יכל להיות הרבה באצ'ים.

productCategory- לכל מוצר מקוטלג על פי שלושה קטגוריות כך שכל קטגוריה היא תת קטגוריה של השנייה.

discounts - הנחה על מוצר מסויים או קטגוריה אך לא על שניהם ביחד מאופיין עם תיאור קטן על הנחה כמות אחוזים וטווח תאריכים לאותה הנחה.